

VEERABABU SUTAPALLI

Machine Learning & GenAI Engineer

Hyderabad, India | isutapalliveerababu@gmail.com | Portfolio

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Professional Summary

Computer Science postgraduate focused on machine learning, GenAI, LLM systems, and practical AI application development. Experienced in distributed LLM inference research, AI/LLM product-integration support, large-scale data processing, and ML/NLP projects using Python, PyTorch, Hugging Face, Scikit-learn, and PySpark. T-shaped engineer with supporting strengths across data engineering, backend APIs, full-stack systems, and cloud-backed delivery.

Technical Skills

AI & Machine Learning	Python, pandas, NumPy, Scikit-learn, PyTorch Basics, Supervised Learning, Feature Engineering, Cross Validation, Model Evaluation, Explainable AI
GenAI & NLP	Hugging Face Transformers, LLM Inference, NLP, Text Classification, Prompt Engineering, Embeddings, RAG Basics, Structured Outputs, LIME
Data Engineering	SQL, MySQL, PySpark, HDFS, ETL, Distributed Processing, Data Cleaning, Parquet
Backend & Full-Stack	Flask, Django, React, Next.js, JavaScript, TypeScript, REST APIs, Firebase
Cloud, Testing & Tools	Azure VMs, AWS, Docker, Git, GitHub, Postman, Playwright, Selenium, Jupyter/Colab

Experience

Nordic Stone Studio

Programming Intern

Malmö, Sweden

Feb 2024 – Jun 2024

- Contributed to Tile Cozy's editor and product workflows by quickly understanding an existing Python, JavaScript, Django, and React codebase and implementing application features.
- Supported LLM and Generative AI integration into product systems by preparing data inputs, contributing implementation support, and prototyping AI-assisted workflow components.
- Collaborated through team meetings, technical discussions, and Git/GitHub workflows while consistently completing assigned development tasks within internship timelines.

JNTU Incubation Centre, Innovation Research Centre

Big Data and Machine Learning Engineer Intern

Kakinada, India

Jan 2022 – Jul 2022

- Reduced computation time by 30% while processing 100M+ healthcare and insurance records by engineering PySpark-based distributed workflows over HDFS.
- Improved data consistency and reduced project time by 20% by building ETL pipelines for patient, hospital, and insurance-claim analytics.
- Supported insurance-eligibility prediction and web-based result delivery by preparing ML-ready datasets and collaborating on Python/MySQL-backed implementation.

JNTU Incubation Centre, Innovation Research Centre

Mobile Application Developer Intern

Kakinada, India

Jun 2021 – Dec 2021

- Developed Python/Flask REST API and MySQL-backed workflows for the Agricultural Advisory Board application, supporting category-wise digitization of farmer grievances.
- Reduced analysis time by 40% through MySQL query optimization and improved reliability using unit checks, Selenium automation, and Postman API validation.

Selected Projects

AI-Generated Review & Comment Detection

GitHub

Python, NLP, Random Forest, Pseudo-Labeling, Explainable AI

- Built text-classification workflows across 487,898 labelled and 443,079 unlabelled records using cleaning, normalization, lemmatization, tokenization, EDA, and pseudo-labelling experiments.
- Evaluated baseline, pseudo-labelled, and tuned Random Forest models using accuracy, precision, recall, F1-score, and AUC-ROC; achieved 0.85 accuracy and 0.94 AUC-ROC after tuning.
- Improved model transparency through LIME, feature-importance analysis, and partial dependence plots.

Decision Support System for Port Efficiency Analysis

GitHub

Python, Flask, pandas, Data Analytics, ML Regression, Visualization

- Contributed to a Flask-based decision-support system analysing 1,330,169 maritime-operation records across 25 columns, including transit, waiting, dwell, vessel-speed, and source-destination patterns.
- Compared Linear Regression, Decision Tree Regressor, and Gradient Boosting Regressor using MAE, MSE, and R-squared, and translated findings into stakeholder-facing analytics views.

Android Malware Detection using Machine Learning

GitHub

Python, Scikit-learn, SVM, Random Forest, KNN, Cybersecurity

- Compared Decision Tree, Random Forest, SVM, and KNN classifiers for Android malware detection using accuracy, precision, recall, F1-score, AUC-ROC, and statistical significance testing.
- Identified SVM as the strongest overall model, reaching 97.11% accuracy and leading F1-score and AUC-ROC performance, while Random Forest achieved the highest precision.

Cloud-Based Brain Stroke Detection

GitHub

Python, Flask, Docker, AWS ECR/ECS, Cloud Monitoring

- Developed and containerized a Flask-based brain-stroke detection application, pushed the Docker image to AWS ECR, and deployed it through AWS ECS.
- Monitored CPU and memory utilisation through AWS metrics and documented the end-to-end cloud deployment workflow.

Research & Publications

Scaling Distributed Inference of BLOOMZ-560M Lightweight Large Language Models using Petals Tool Publication

Master's Thesis, Blekinge Institute of Technology, 2026

- Analysed distributed LLM inference across 1–10 Azure VM nodes using 42,989 prompts and sequential, pipelined, and microbatched execution strategies.
- Identified pipelined execution as the strongest scaling strategy with approximately $1.6\times$ mean speedup at 10 nodes, while observing a $13\times$ increase in network footprint.

Predicting Risk Level in Life Insurance Application

Publication

Bachelor's Thesis, Blekinge Institute of Technology, 2023

- Compared Logistic Regression, Decision Tree, Random Forest, and Linear SVC on unmodified and modified life-insurance datasets using accuracy, precision, recall, and F1-score.
- Identified Random Forest as the strongest classifier with 53.79% accuracy on the unmodified dataset and 52.80% on the modified dataset.

Education

Blekinge Institute of Technology

Karlskrona, Sweden

Master of Science in Computer Science – 50% tuition scholarship based on academic merit

Aug 2023 – May 2026

Jawaharlal Nehru Technological University Kakinada

Kakinada, India

Bachelor of Technology in Computer Science and Engineering

Aug 2019 – Jun 2023

Selected Certifications

Machine Learning

Coursera

Programming, Data Structures and Algorithms using Python

NPTEL

Python (Basic) Skill Assessment

HackerRank

Machine Learning to Deep Learning: Remote Sensing Data Classification

IIRS / ISRO

Honors & Achievements

- Secured 2nd rank at college level in the Scaler Edge Apex 2021 Hackathon.
- Earned 6-star HackerRank recognition in Problem Solving and Python Programming.